ExamNexus

AI-Assisted Examination Resource Orchestration

From Examination Management to Adaptive Resource Coordination

TechGig — Sovereign Technology for India 2026

Artificial Intelligence & Deep Tech
Plan Better. Allocate Smarter. Adapt Faster.

Team
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The Examination Planning Problem

University examinations involve interdependent resources that must be coordinated together.

Why it is difficult

A change in one resource can affect multiple arrangements.

Core Question

How can universities create valid examination plans while responding to changing conditions — without losing control of critical decisions?

The Challenge

Students

Eligibility & accessibility

Examinations

Schedules & conflicts

Halls

Capacity & availability

Staff

Invigilation & workload

The ExamNexus Approach

One connected workflow for examination resource orchestration

Institutional Data

Requirements & Constraints

Allocation & Optimization

Validation

Administrator Approval

Execution

ExamNexus connects

- Examinations
- Students & Invigilators
- Halls
- Seats
- Accessibility

Core principle

- AI assists.
- Deterministic rules validate.
- Administrators remain in control.

WORKING PROTOTYPE

A functional examination management platform connecting institutional data, planning, allocation and execution.

WHAT IS IMPLEMENTED

Core Management

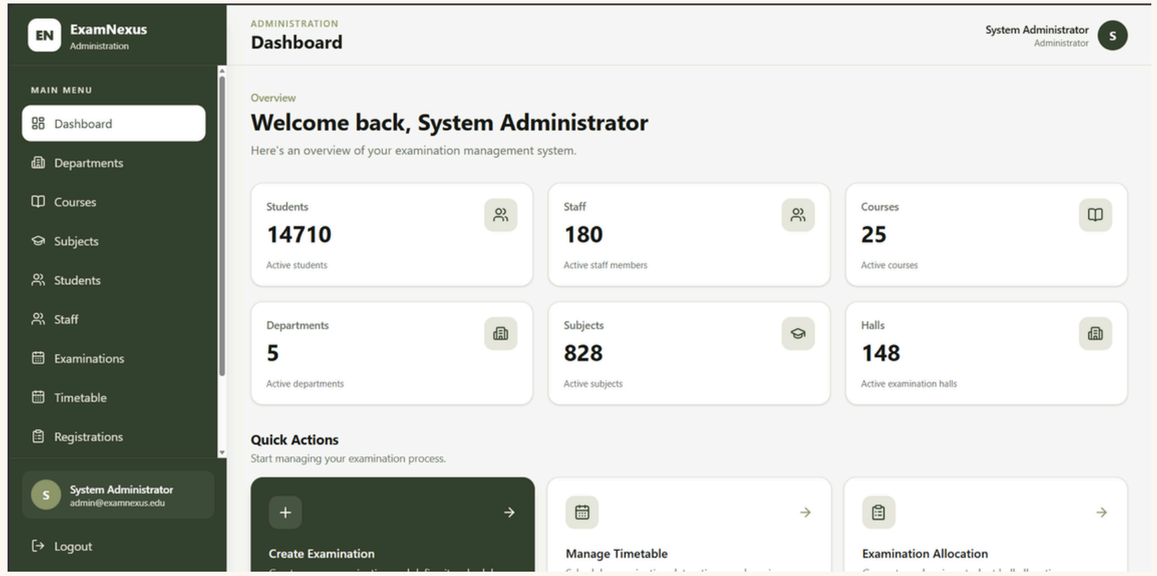
- Students & staff
- Departments & courses
- Subjects & examinations
- Examination halls & resources

Decision Support

- Constraint validation
- Conflict detection
- Optimization
- Administrative review & approval

Examination Operations

- Timetable & eligibility
- Hall allocation
- Accessibility-aware allocation
- Seat allocation
- Invigilator allocation
- QR hall tickets

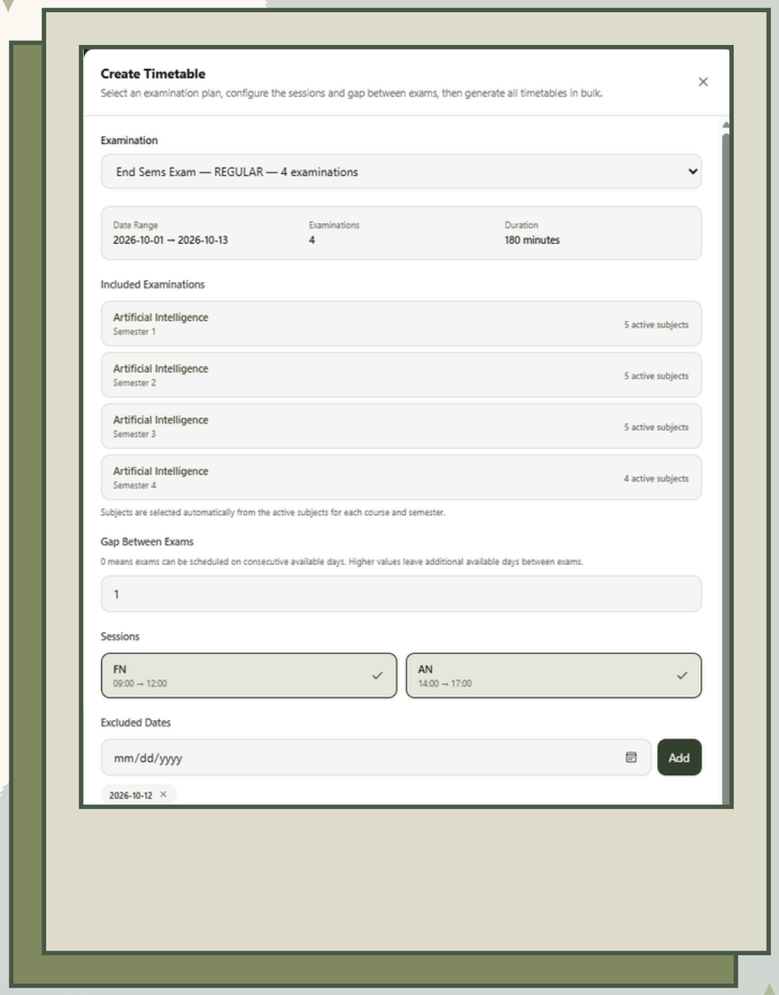


Live Prototype Dashboard — Current Dataset

14,710 Students | 180 Staff | 25 Courses | 5 Departments | 828 Subjects | 148 Halls

FROM INSTITUTIONAL DATA TO EXAMINATION EXECUTION

MANAGE DATA	Students • Staff • Departments • Courses • Subjects • Halls
CREATE EXAMINATION	Examination • Subjects • Schedule • Eligibility
PREPARE RESOURCES	Hall availability • Capacity • Accessibility • Staff
ALLOCATE	Halls • Seats • Invigilators
VALIDATE & OPTIMIZE	Constraints • Conflicts • Workload • Resource utilization
REVIEW & APPROVE	Administrator Reviews the Final Arrangement
EXECUTE	Hall Tickets • QR Verification • Reports • Examination Operations



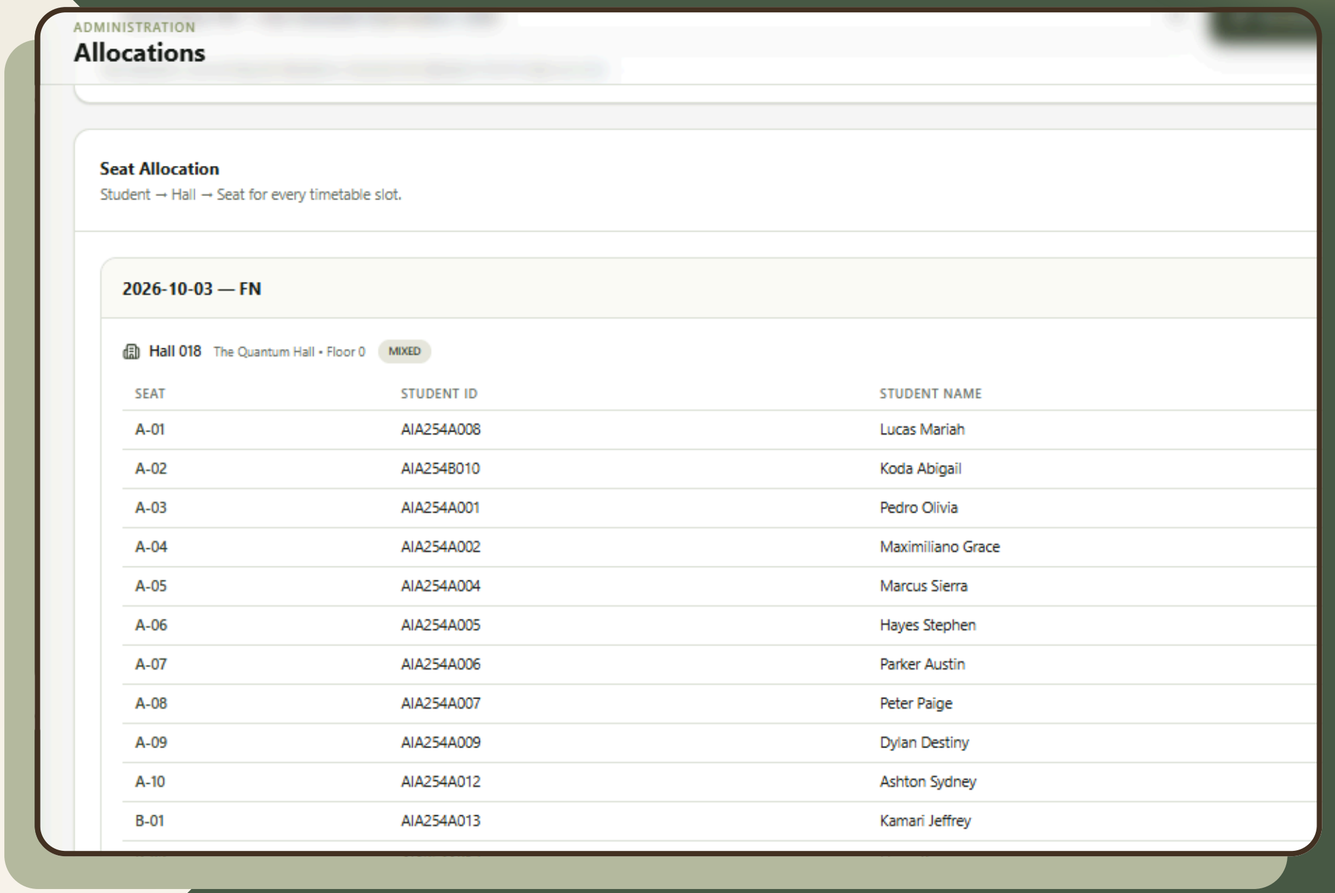
INTELLIGENT EXAMINATION ALLOCATION

HARD CONSTRAINTS

- Student eligibility
- Hall capacity
- Hall availability
- Examination conflicts
- Accessibility requirements
- Staff availability

OPTIMIZATION

- Better hall utilization
- Fewer halls where possible
- Balanced invigilator workload
- Reduced unnecessary movement



DECISION FLOW

VALID FIRST → OPTIMIZE → VALIDATE → APPROVE

PROTOTYPE

CONSTRAINT-AWARE EXAMINATION ALLOCATION

AI-ASSISTED EXAMINATION DECISION SUPPORT

AI SUPPORTS THE ADMINISTRATOR ACROSS THE EXAMINATION WORKFLOW

UNDERSTAND

Natural-language requirements

Administrator describes examination requirements in plain language.

QUERY

Natural-language questions

Administrators can ask questions about examination data and resources.

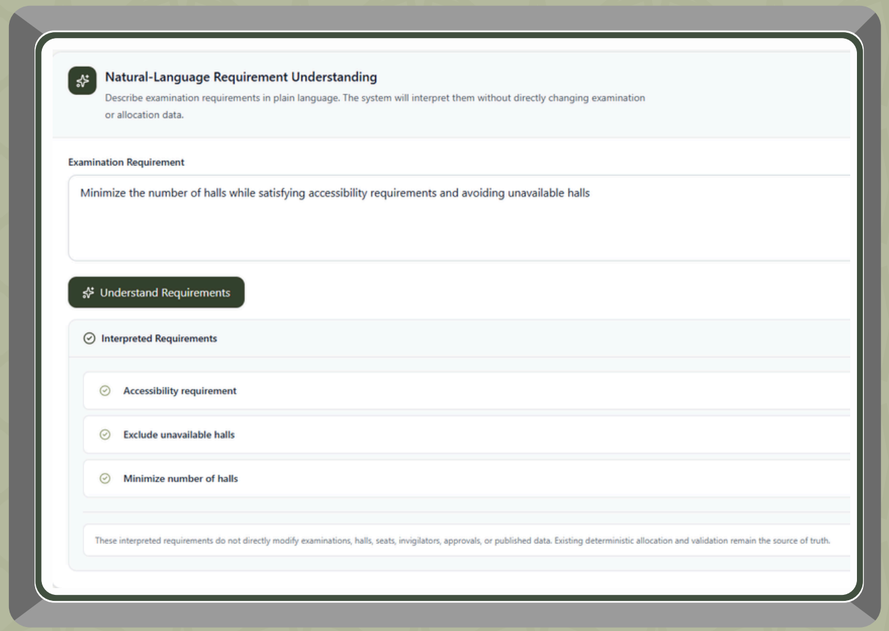
EXPLAIN

Understand allocation decisions

AI helps explain decisions using actual examination and allocation information.

ARCHITECTURE

- ADMINISTRATOR
- NATURAL LANGUAGE
- AI ASSISTANCE
- STRUCTURED / VALIDATED REQUEST
- EXAMNEXUS DECISION SERVICES
- VALIDATION
- RESULT



WORKING PROTOTYPE

AI-ASSISTED REQUIREMENT / QUERY

DESIGN PRINCIPLE

AI-assisted. Human-controlled.

AI supports decision-making without bypassing validation or administrator approval.

AI-ASSISTED. DETERMINISTIC AT THE CORE.

WHY THIS ARCHITECTURE?

AI provides natural-language interaction and decision support.

AI Assistance Layer

→ Understand
→ Interpret
→ Query
→ Explain

ExamNexus Decision Layer

→ Constraints
→ Allocation
→ Optimization
→ Validation

Human Control

→ Review
→ Modify
→ Approve

CRITICAL EXAMINATION DECISIONS REMAIN GOVERNED BY:

- Explicit Rules
- Validated Data
- Deterministic Allocation Logic
- Constraint Checking
- Administrator Approval

KEY PRINCIPLE

AI adds intelligence without making the examination system uncontrolled.

ACCESSIBILITY-AWARE ALLOCATION

Accessibility is built into the allocation process — not treated as an afterthought.

Student Requirement

Eligible Resources

Accessibility Check

Capacity & Availability

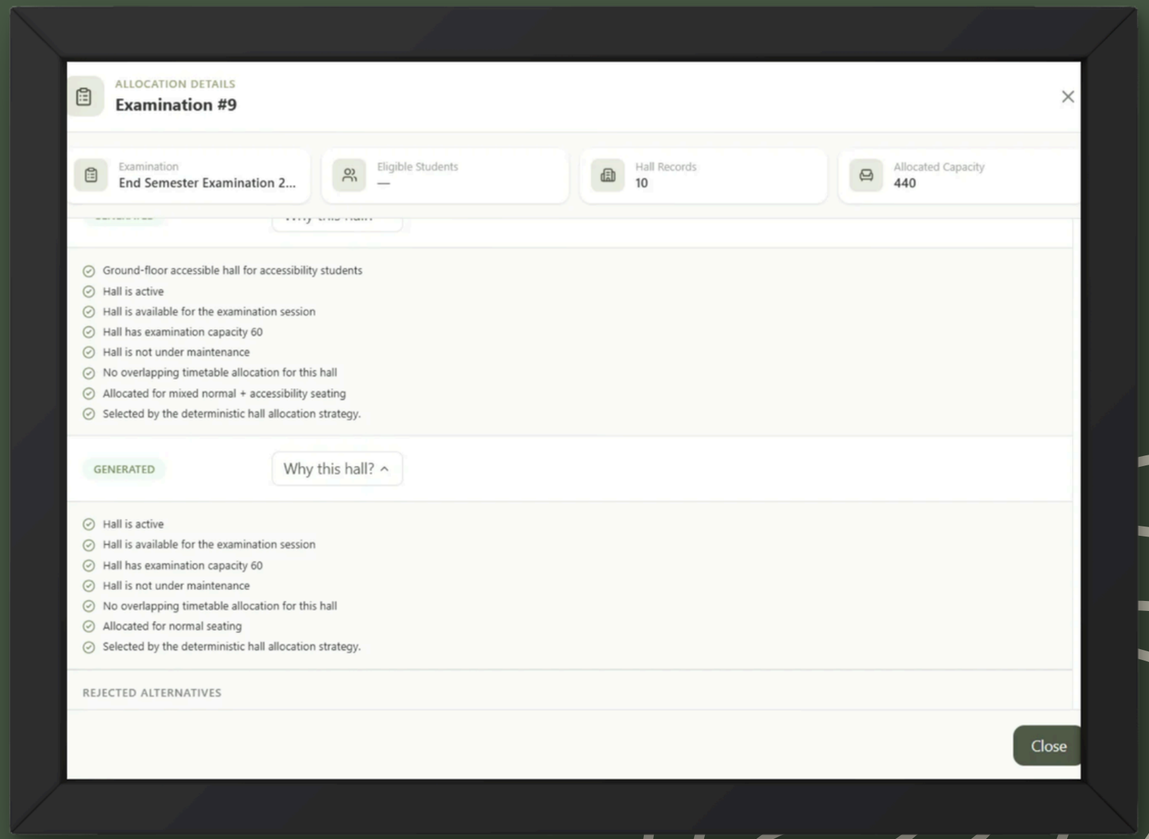
Allocation

Validation

EXAMPLE

Accessibility requirement identified

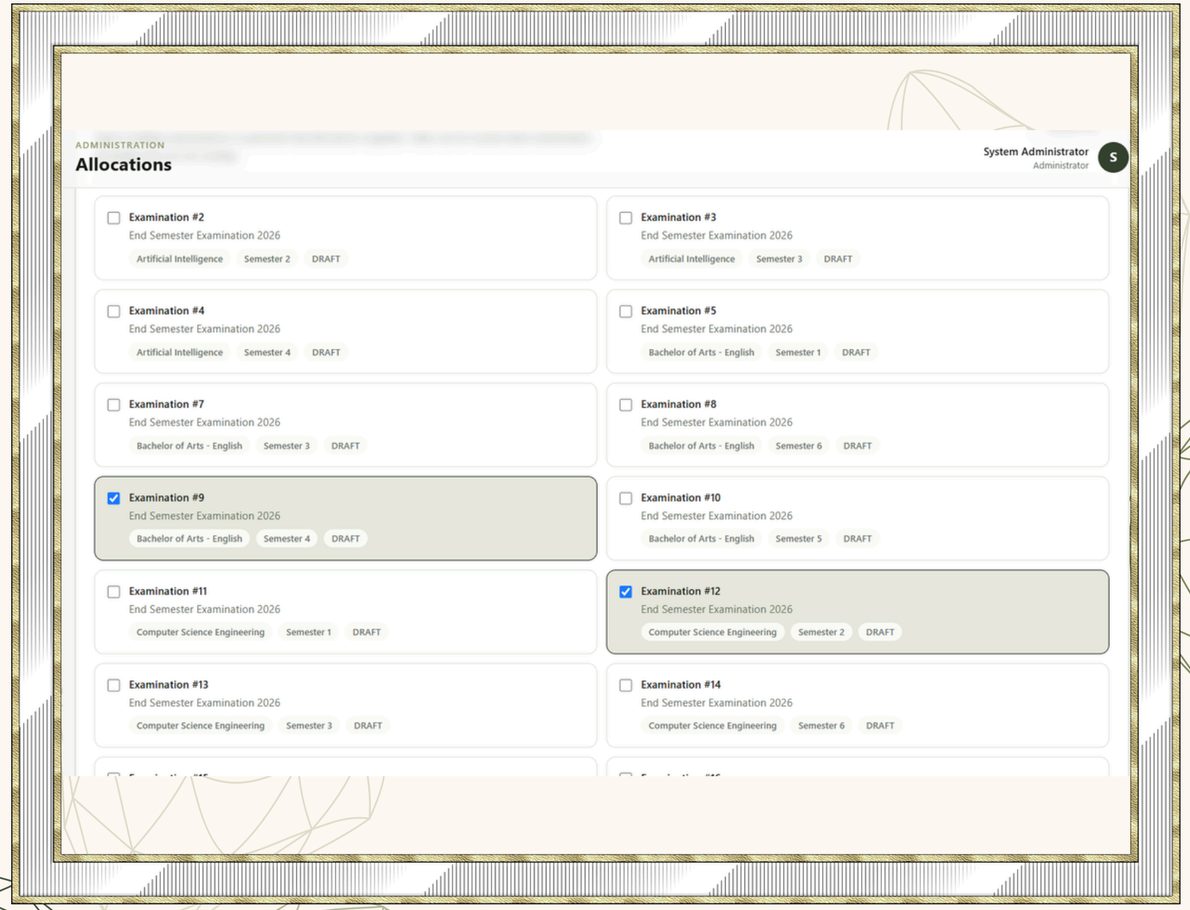
- Suitable hall considered
- Capacity verified
- Availability verified
- Final allocation validated



INCLUSION BY DESIGN

Accessibility becomes an allocation constraint, not a manual exception.

CONNECTED EXAMINATION RESOURCE ALLOCATION



One Connected Examination Plan

Students + Halls + Seats + Invigilators

HALL ALLOCATION

Students → Suitable Halls

SEAT ALLOCATION

Students → Specific Seats

INVIGILATOR ALLOCATION

Staff → Examination Duties

VALIDATION ACROSS RESOURCES

- Capacity
- Availability
- Eligibility
- Conflicts
- Accessibility
- Workload

ADAPTIVE EXAMINATION PLANNING

When examination conditions change, ExamNexus can replan.

Example — Hall H204 Becomes Unavailable

ADAPTIVE LOOP

Plan → Detect Change → Re-plan → Validate → Review → Approve → Adapt

01 — DETECT

Hall becomes unavailable

02 — IDENTIFY IMPACT

Affected arrangements are identified

03 — FIND ALTERNATIVES

Available resources are considered

04 — REPLAN

Alternative arrangements are generated

05 — VALIDATE

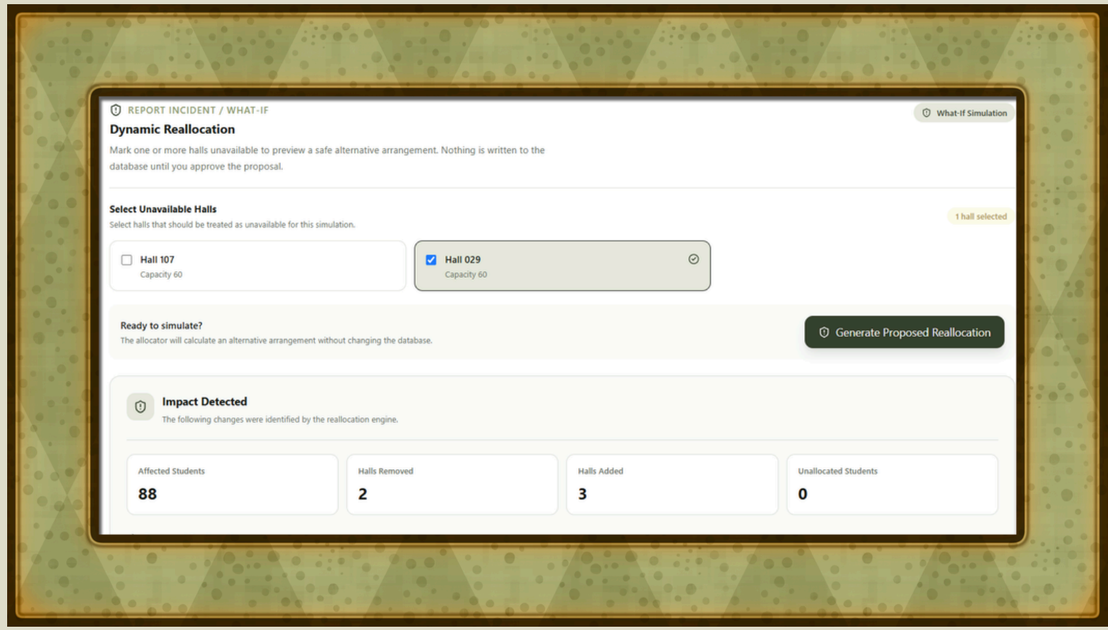
Capacity • Schedule • Accessibility • Availability

06 — REVIEW

Impact and alternatives are presented

07 — APPROVE & APPLY

Administrator controls the final change



KEY PRINCIPLE

Simulate → Validate → Review → Approve → Apply

VALIDATION & PROTOTYPE EVIDENCE

Before execution, ExamNexus independently validates the examination arrangement.

VERIFIED · SUCCESSFUL

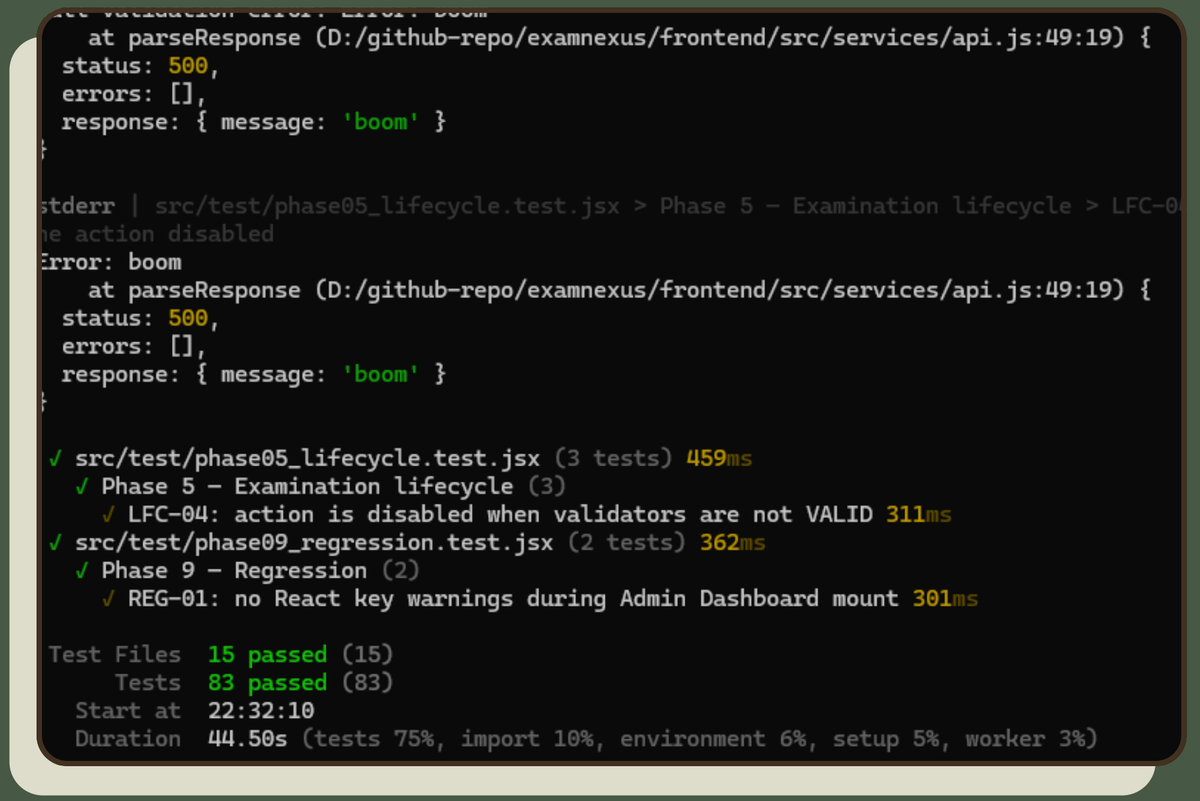
449	Backend Tests Passed
83	Frontend Tests Passed
6	Backend Tests Skipped
0	Backend Tests Failed

Independent Validation Checks

- Eligibility & Allocation Completeness
- Hall Capacity & Availability
- Timetable & Resource Conflicts
- Accessibility Requirements
- Invigilator Workload

REVIEW FLOW

Generate → Validate → Review → Approve → Execute



VERIFIED · AUTOMATED TESTS SUCCESSFUL

Evidence-based validation of implemented prototype workflows.

Independent Validation Checks
- Eligibility & Allocation Completeness
- Hall Capacity & Availability
- Timetable & Resource Conflicts
- Accessibility Requirements
- Invigilator Workload

Automated Test Evidence

449	83	6	0
Backend Tests Passed	Frontend Tests Passed	Backend Tests Skipped	Backend Tests Failed

VERIFIED · AUTOMATED TESTS SUCCESSFUL

Evidence-based validation of implemented prototype workflows.

Latest recorded results: 14 October 2026. Browser-level and production verification remain separate and are not implied by these test results.

CURRENT PROTOTYPE & NEXT EVOLUTION

CURRENT FUNCTIONALITY

✓ Examination Management
✓ Hall & Resource Allocation
✓ Accessibility-Aware Allocation
✓ Seat & Invigilator Allocation
✓ Constraint Validation
✓ Optimization
✓ QR Hall Tickets
✓ AI-Assisted Requirements & Queries
✓ Decision Explanation
✓ Dynamic Reallocation
✓ Intelligent Data Validation

FUTURE — NEXT DIRECTIONS

- Predictive Resource Planning
- Advanced Institutional Analytics

ExamNexus

Plan Better. Allocate Smarter. Adapt Faster.

AI-assisted examination resource orchestration for connected, constraint-aware and adaptive examination planning.

AI-Assisted • Constraint-Aware • Optimized • Adaptive • Human-Controlled

THANK YOU

TEAM

MARIA DORTHY · MARIA SILVIA

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